



CANDIDATE  
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**ANSWER KEY**

CIVICS  
GROUP

**2**

**1**

**-**

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## H2 Biology

**9744/02**

Paper 2 Structured Questions

**14 September 2022**

**2 hours**

Candidates answer on the Question Paper.  
No Additional Materials are required.

### READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

#### For Examiner's Use

|              |            |
|--------------|------------|
| <b>1</b>     |            |
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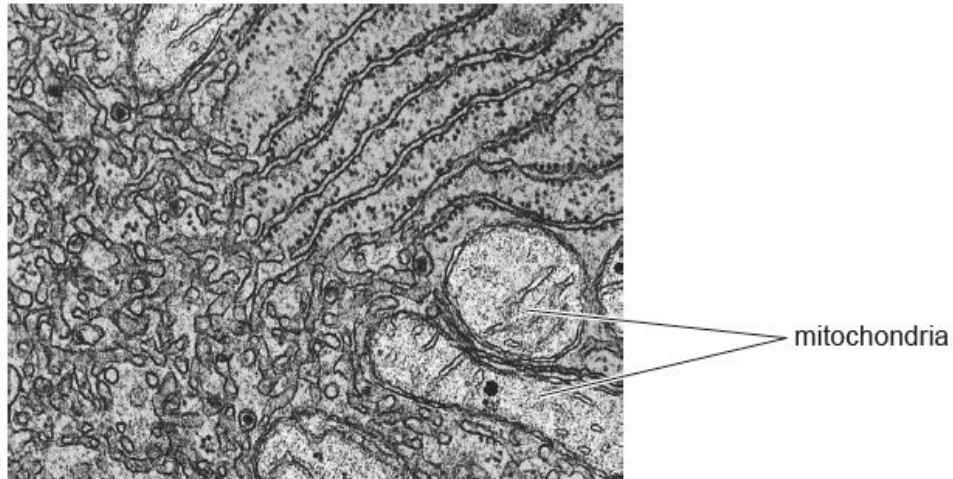
This document consists of **30** printed pages and **2** blank pages.

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Answer **all** questions.

- 1 The Golgi body, rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) form part of the internal membrane system of a cell. The membranes have a fluid mosaic structure.

Fig. 1.1 is a transmission electron micrograph of one area of a liver cell showing a region with RER and a region with SER. Mitochondria are also visible in the image.



**Fig. 1.1**

- (a) Describe **one** difference between RER and SER. [2]

Structure (any 1)

1. RER has ribosomes attached to its surface but SER has **no** ribosomes attached;
2. RER has flattened sacs / cisternae but SER is tubular;
3. RER is continuous with **outer membrane of nucleus** (R: nuclear envelope) but SER is not connected to the nucleus;
4. AVP e.g. RER has **regular / layered** arrangement compared to SER which has **irregular / disorganised** arrangement;

Function

5. RER produces / transports, proteins / glycoproteins / polypeptides while SER produces lipids / cholesterol / steroids; (A: detoxification)

- (b) Phospholipids are one of the main components of membranes.

Describe the structure of a phospholipid molecule. [2]

1. A phospholipid consists of 1 phosphoric acid / phosphate group and 2 fatty acids combined with 1 glycerol; (R: if quantity of components not stated)
2. Each fatty acid forms an ester bond with glycerol (total of 2 ester bonds) and phosphoric acid forms a phosphoester bond with glycerol; (NOTE: glycerol for point 1 can be marked here if student wrote phosphate head and 2 fatty acid tails but did not mention glycerol as part of the structure)
3. AVP e.g. Choline portion attached;
4. AVP e.g. Saturated / unsaturated fatty acid tails;

Point 1 must be present for full credit.

(c) One function of a Golgi body is to package molecules into Golgi vesicles.

(i) A Golgi body and Golgi vesicles are **not** visible in Fig. 1.1.

Describe the features, **other than** the presence of Golgi vesicles, that would help you identify a Golgi body in a transmission electron micrograph of another area of the same liver cell. [2]

1. Stack of flattened membrane-bound sacs called cisternae;
2. Each cisterna in stack is **not interconnected** / **separate**;
3. Has a single membrane;
4. Two distinct faces (**cis and trans**);
5. Two ends of cisternae are swollen / enlarged; (max 2)

Point 5 is not recommended (not technical).

(ii) Some Golgi vesicles contain secretory proteins for release from the cell.

Describe the sequence of events that occurs following the packaging of a secretory protein into a Golgi vesicle to its release from the cell. [3]

1. In Golgi apparatus, secretory proteins leave via **secretory vesicles / Golgi vesicles** by budding from **trans** face of Golgi;
2. Secretory vesicles / Golgi vesicles containing secretory proteins fuse with cell surface membrane;
3. and **release** secretory proteins via exocytosis; (R?: proteins *exit/leave* cell) (I: all events *before packaging of secretory protein into Golgi vesicle*)

(iii) Some Golgi vesicles contain glycoproteins or glycolipids to be added to the cell surface membrane.

State a role of **glycolipids** in the cell surface membrane. [1]

1. Acts as a **self-antigen / recognition site** / for **cell-cell recognition**;
2. Acts as a **receptor**;
3. For **membrane stability**;
4. For **cell-cell adhesion**; (any 1)

[Total: 10]

- 2 Fig. 2.1A shows a polypeptide molecule during protein synthesis. A molecule of glycine is shown just before it is added to the polypeptide.

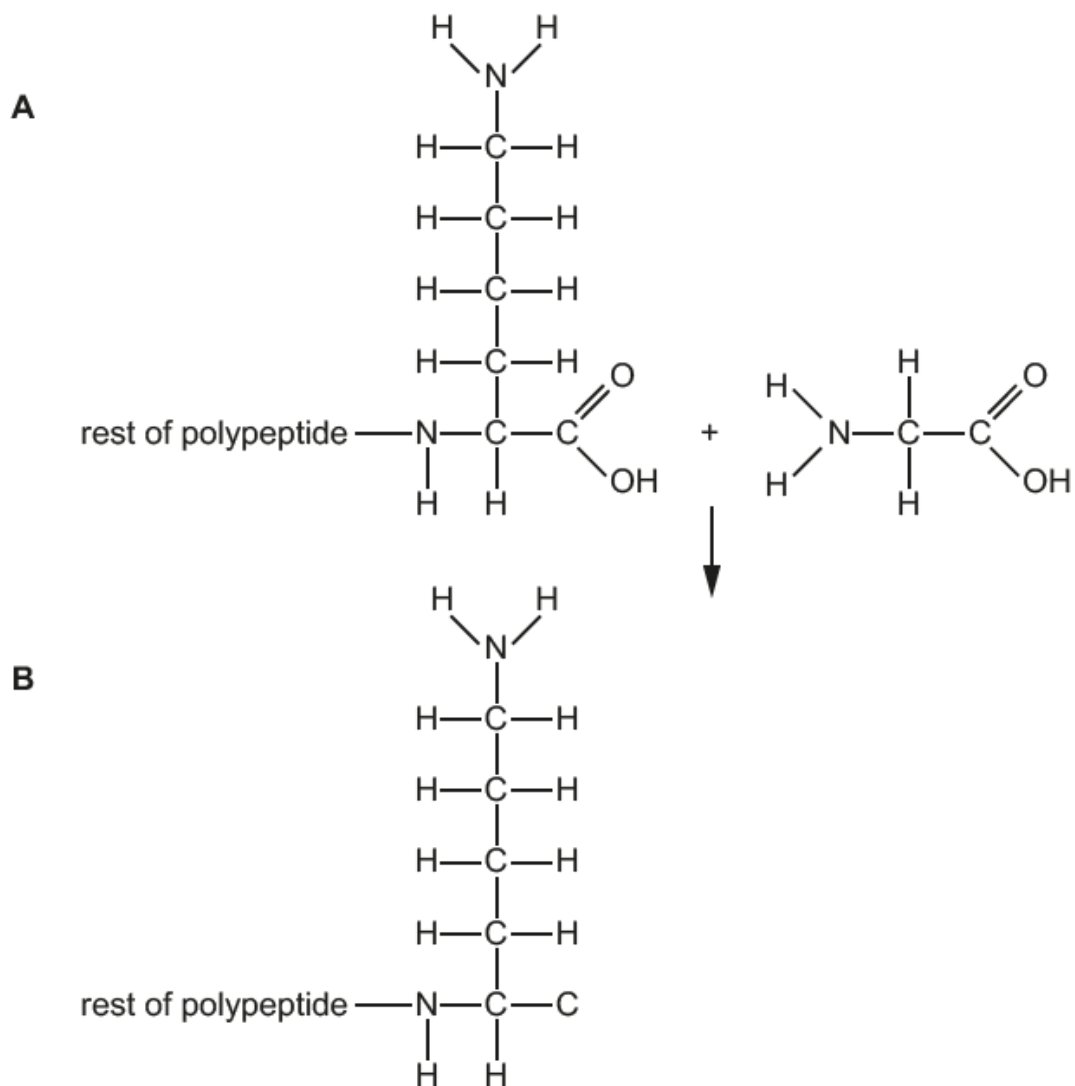


Fig. 2.1

(a) Complete Fig. 2.1B to show the molecule of glycine added to the end of the polypeptide. [2]

1. **Product** (R: R used instead of H for glycine) + release of **water** molecule
2. Peptide **bond**: correct identification + labelling

(Not marked for but should include: condensation **reaction**)

The feet of elephants are protected by structures under the skin known as cushions, as shown in Fig. 2.2.



Fig. 2.2

The cushions are made up of a large number of cells surrounded by connective tissue containing many fibres of collagen. The fibres help to maintain the structure of the cushion.

The collagen fibres are made of collagen molecules.



Fig. 2.3

With reference to Fig. 2.3,

(i) describe the structure of a wild-type collagen molecule. [4]

1. **Three polypeptide chains coil around each other** to form a **tight/compact triple helix**;
2. **Each polypeptide chain** consists of **repeating tripeptide units: G-X-Y**, where **G** is glycine, **X** is usually proline and **Y** is usually hydroxyproline; (glycine is every third amino acid in the polypeptide chain) (R: just glycine, proline, hydroxyproline)
3. **Hydrogen bonds** are formed **between adjacent polypeptide chains**, specifically between peptide bond **N-H group** of **glycine** and **C=O group** of **proline**
4. **G** is **smaller than X and Y**, which are **bulkier and less flexible**

(ii) Osteogenesis imperfecta (OI), commonly known as "brittle bone disease", is a disorder characterized by bone fragility and abnormalities of connective tissue. Vast majority of affected individuals have mutations in the genes encoding the polypeptide chains of collagen, which result in substitution of amino acid residues.

Suggest the effect of such mutations on the structure of collagen fibre. [2]

1. **G** is substituted with a **much larger amino acid residue** which **disrupts the close/compact packing of triple helix / three polypeptides** ; (R: no collagen molecule/tropocollagen formed) [compulsory]

2. Less/no covalent cross-links between adjacent collagen molecules, reducing formation of collagen fibrils, and disrupting structure of collagen fibre formed from staggered arrangement of collagen fibrils;

Collagen is a water-insoluble fibrous protein that can be prone to swelling. Collagen swelling occurs when water fills up the gaps between collagen molecules.

Fig. 2.4 shows the effect of pH on swelling ratio of collagen. Swelling ratio is defined as the percentage increase in the weight of collagen due to water absorption.

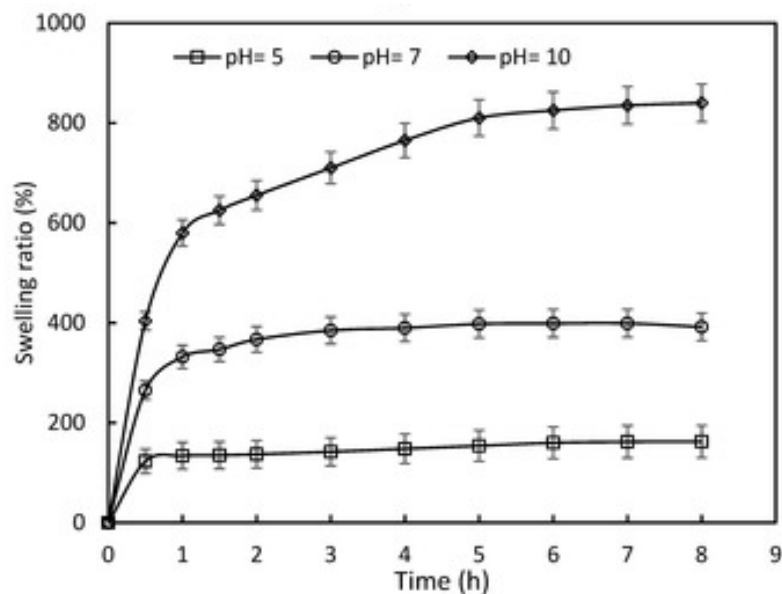


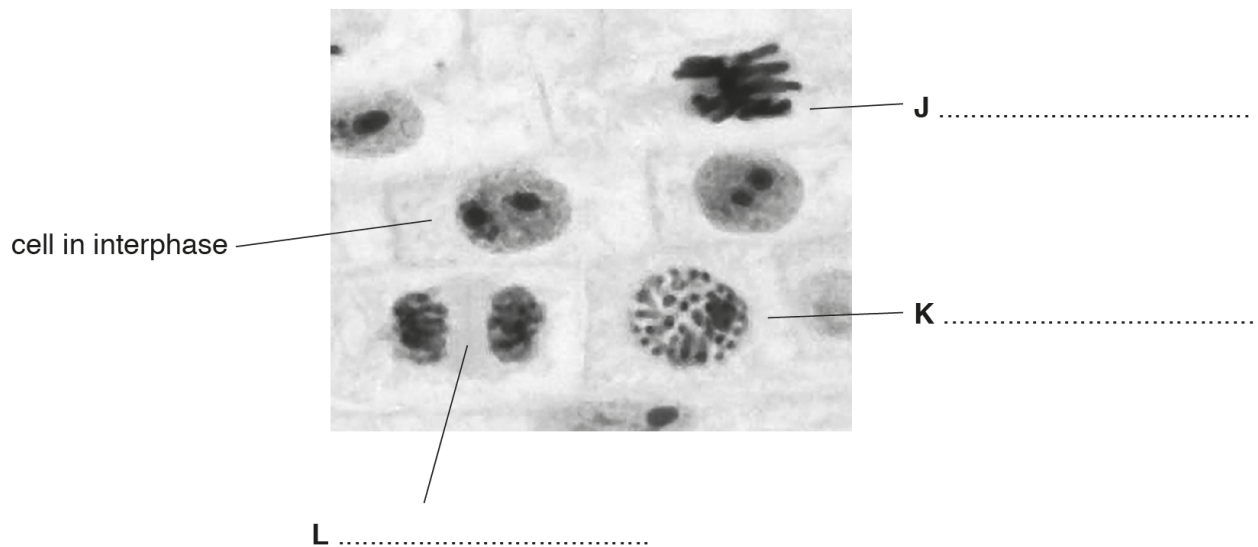
Fig. 2.4

(iii) With reference to Fig. 2.4, describe and explain the effect of pH on collagen. [2]

1. As pH increases from pH 5 to pH 7 to pH 10, the swelling ratio of collagen increases from around 150% to 400% to 850% by end of 8 hours;
2. Excess OH<sup>-</sup> ions at high pH neutralise the positively charged R group of amino acids, breaking the ionic bonds and hydrogen bonds, creating larger / more gaps between collagen molecules, resulting in higher swelling ratio;

[Total: 10]

- 3 (a) Fig. 3.1 is a photomicrograph of root tip cells at different stages in the cell cycle. A cell cycle in interphase is labelled.



**Fig. 3.1**

- (i) Name the stage of mitosis shown in each of the cells **J**, **K** and **L** in Fig. 3.1.

Write your answer in the space next to each letter on Fig. 3.1.

[2]

1. **J** metaphase; **K** prophase; **L** telophase  
[3 correct: 2 marks; 1 – 2 correct: 1 mark]

- (ii) Explain how it is possible to deduce that the labelled cell in interphase shown in Fig. 3.1 is in late, rather than early, interphase. [1]

1. The size of the cell is the same size as cells in mitosis (OWTTE)

- (iii) Suggest one advantage of using a light microscope, rather than an electron microscope, to study cell division. [1]

1. Light microscope can be used to observe living cells
2. the process of mitosis can be seen happening (OWTTE) (**R** easier to see mitosis happening **I** to study cell division)
3. The light microscope is easier to use (OWTTE) A: qualified personnel needed for use of electron microscope.
4. AVP

(b) Mutations in human body cells can sometimes result in a tumour. Some tumours are cancerous.

(i) Outline how tumour development is a multistep process. [3]

1. A single normal cell first undergoes a loss-of-function mutation in both alleles of the tumour suppressor gene, leading to increased cell proliferation/excessive cell division;
2. Following this, the cell accumulates another gain-of-function mutation of the proto-oncogene to form oncogene,
3. Coupled with other accumulated mutations in other cancer critical genes (e.g. proto-oncogenes and telomerase genes), they lead to uncontrolled cell division and the development of tumour.
4. AVP: **Dysregulation of cell cycle checkpoints** due to mutations.

(ii) Tumour cells have antigens on their cell surface that are not present on non-tumour cells.

These antigens are the result of gene mutations and are known as tumour specific antigens (TSA).

One type of TSA differs in structure from the protein found on the cell surface of non-tumour cells by a single amino acid.

Explain how a gene mutation could result in the production of this TSA. [2]

1. It is due to a single base substitution mutation, which results in a different mRNA codon that codes for different amino acid. (A: description of single base mutation, e.g. point mutation)
2. Thus, the resulting protein will have a different primary structure / the sequence of amino acids is changed. Subsequently changing the secondary and tertiary structure, causing polypeptide to fold into a different 3D conformation (A: tertiary structure).

(c) Immunotherapy is a form of treatment for cancer which aims to stimulate the immune system to destroy tumour cells.

One form of immunotherapy for cancer uses a vaccine which contains one specific type of TSA.

Suggest how vaccination with a specific type of TSA could lead to the destruction of tumour cells by T-lymphocytes in the body. [1]

1. **Activated T-lymphocytes** will **only recognize** one specific type of **TSA on tumour cells** and destroy tumour cells only. (OWTTE)

I: antigen presentation, antibody production

[Total: 10]

- 4 Lactate dehydrogenase (LDH) is an enzyme found in many organisms. Within the same organism, it can be found in different forms, called isoenzymes. The isoenzymes are structurally different but all catalyse the same reaction.

Lactate dehydrogenase isoenzymes are globular proteins, each consisting of four polypeptides.

Lactate dehydrogenase isoenzymes are made up of two types of polypeptide: polypeptide M, which is coded for by the LDH-A gene and polypeptide H, which is coded for by the LDH-B gene.

Table 4.1 shows the composition of different human lactate dehydrogenase isoenzymes and examples of tissues and organs where each can be found.

**Table 4.1**

| isoenzyme | polypeptide composition of enzyme | example of isoenzyme location |
|-----------|-----------------------------------|-------------------------------|
| LDH-1     | HHHH                              | heart<br>red blood cells      |
| LDH-2     | HHHM                              | heart<br>red blood cells      |
| LDH-3     | HHMM                              | brain<br>lungs                |
| LDH-4     | HMMM                              | kidneys<br>placenta           |
| LDH-5     | MMMM                              | liver<br>skeletal muscles     |

- (a) With reference to Table 4.1, suggest how different cells of the same individual can produce different isoenzymes. [2]

1. Due to differential gene expression / WTTE of LDH-A and LDH-B genes expressed in different cells; (A: Appropriate reference to mRNA half-life of LDH-A and LDH-B mRNA products)

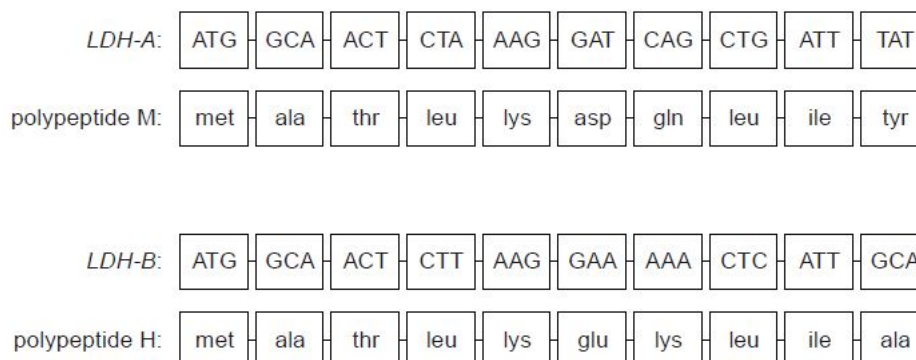
(R: reference to alternative splicing since there are only two genes given, each gene only codes for one polypeptide – LDH-A gene for polypeptide M and LDH-B for polypeptide H, hence not likely to have alternative splicing)

2. Cite appropriate pair of data from Table 4.1 e.g. LDH-1 isozyme from heart and red blood cells have only H polypeptide chains suggesting only LDH-B gene expressed but LDH-5 isozyme in liver and skeletal muscles only has 4 M polypeptide chains, suggesting that only LDH-A gene expressed;  
(LDH-1 and LDH-5 is a straightforward pair for use to compare)

Explanation given must match data cited e.g. genes being switched off cannot be used to account for presence of LDH-2 to LDH-4 isoenzymes as both the LDH-A and LDH-B would be expressed (they contain mixture of H and M polypeptides).

- (b) The base sequences of the LDH-A and LDH-B genes and the sequences of the amino acids encoded by these genes were determined.

Fig. 4.1 shows the first ten amino acids of polypeptides M and H and the corresponding base sequences of one of the DNA strands of each gene.



**Fig. 4.1**

Table 4.2 shows the genetic code (mRNA codons).

**Table 4.2**

| first position | second position |     |      |      | third position |
|----------------|-----------------|-----|------|------|----------------|
|                | U               | C   | A    | G    |                |
| U              | phe             | ser | tyr  | cys  | U              |
|                | phe             | ser | tyr  | cys  | C              |
|                | leu             | ser | STOP | STOP | A              |
|                | leu             | ser | STOP | trp  | G              |
| C              | leu             | pro | his  | arg  | U              |
|                | leu             | pro | his  | arg  | C              |
|                | leu             | pro | gln  | arg  | A              |
|                | leu             | pro | gln  | arg  | G              |
| A              | ile             | thr | asn  | ser  | U              |
|                | ile             | thr | asn  | ser  | C              |
|                | ile             | thr | lys  | arg  | A              |
|                | met             | thr | lys  | arg  | G              |
| G              | val             | ala | asp  | gly  | U              |
|                | val             | ala | asp  | gly  | C              |
|                | val             | ala | glu  | gly  | A              |
|                | val             | ala | glu  | gly  | G              |

With reference to Fig. 4.1 and Table 4.2,

- (i) state if both the base sequences of LDH-A and LDH-B genes are the template strand or non-template strand of DNA. [1]

Non-template strand(s);

(ii) explain your answer to (b)(i). [2]

1. mRNA is equivalent to DNA strand shown except U replaces T / is complementary copy of transcribed strand of DNA;
2. e.g. Methionine start amino acid thus AUG is mRNA codon, so transcribed DNA would be TAC, instead of ATG;

(iii) Fig. 4.2 shows the process of polypeptide synthesis.

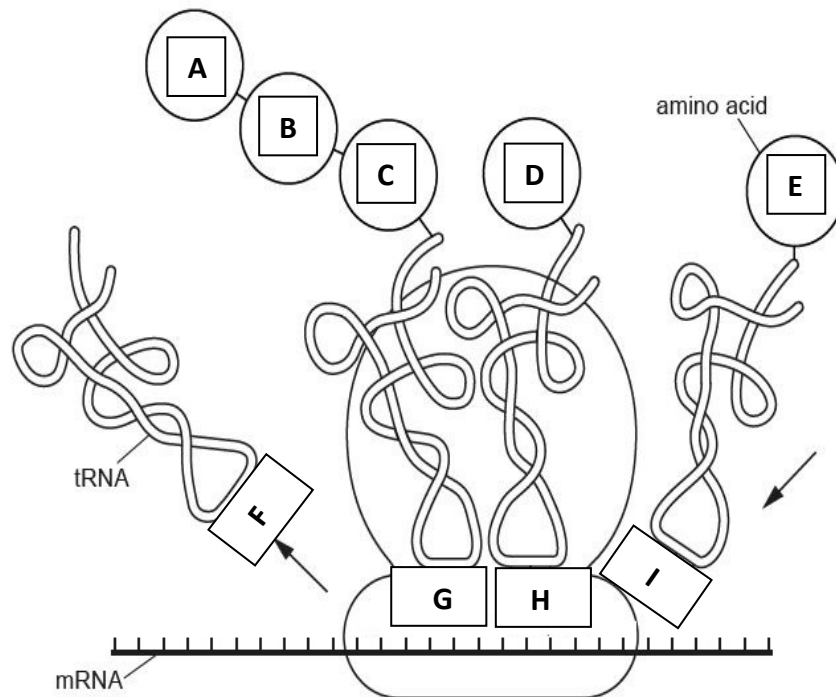


Fig. 4.2

Fill in the following spaces with the relevant information for the synthesis of the first five amino acids for the LDH-A polypeptide chain.

Amino acids: A  B  C  D  E

[1]

Anticodons: F  G  H  I

[1]

CGG      UGG      GAG      UUC  
CGU      UGU      GAU  
CGC      UGC      GAC

1 mark for each group (all or none for each group)

(iv) Compare the structures of mRNA and tRNA. [3]

*Answer to be written in prose. Following table for ease of answer presentation.*

| mRNA                                                                                                                                                                                                                                 | tRNA                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Similarity                                                                                                                                                                                                                           |                                                                    |
| 1. Both are <u>single stranded polynucleotide</u> chain;<br>2. Both are made of <u>ribonucleotides</u> ;<br>3. Both contain <u>uracil</u> in place of thymine;<br>4. AVP; Nucleotides joined by <u>phosphodiester</u> bonds; (max 2) |                                                                    |
| Difference                                                                                                                                                                                                                           |                                                                    |
| 1. Single strand not folded                                                                                                                                                                                                          | 1. Single strand folded into a clover-leaf structure               |
| 2. Absence of hydrogen bonds within molecule                                                                                                                                                                                         | 2. Hydrogen bonds between complementary base pairs within molecule |
| 3. The 3' end does not have a fixed codon                                                                                                                                                                                            | 3. The 3' end of the tRNA strand always end with the CCA codon     |
| 4. No anticodon                                                                                                                                                                                                                      | 4. Anticodon present                                               |
| 5. Does not have amino acid attached                                                                                                                                                                                                 | 5. Can have amino acid attached                                    |
| 6. AVP                                                                                                                                                                                                                               | 6. AVP (Max 2)                                                     |

[Total: 10]

- 5 In multicellular eukaryotes, gene regulation is important in development. Different sets of genes need to be expressed in different cells, at the right times, and in the right sequence for organisms to develop correctly.

In eukaryotes, a specific transcription factor or activator binds to an enhancer to regulate the expression of genes coding for specific proteins in different tissues.

(a) (i) Explain the meaning of the term 'enhancer'. [1]

1. Enhancers are non-coding DNA sequences / distal control elements where **activators bind** to, which increase the rate of transcription;  
(‘Activators bind to’ already suggested in the question stem, but no need to mark for it)

(ii) Using the information given, suggest a reason why activators are referred to as ‘specific’ transcription factors. [1]

1. Specific cell types in the body may have specific types of activator molecules (that are not present in other cell types);
2. AVP; Not found in all eukaryotic species; Only present in specific developmental stage

- (b) The *PITX1* gene codes for a protein that plays a role in the development of pelvic spines of fishes like the three-spine stickleback, *Gasterosteus aculeatus*. *PITX1* also plays critical roles in controlling the development of the fish body like the formation of the jaws, pelvis and pituitary glands.

Fig. 5.1 shows a diagram of the *PITX1* gene and the associated regions.

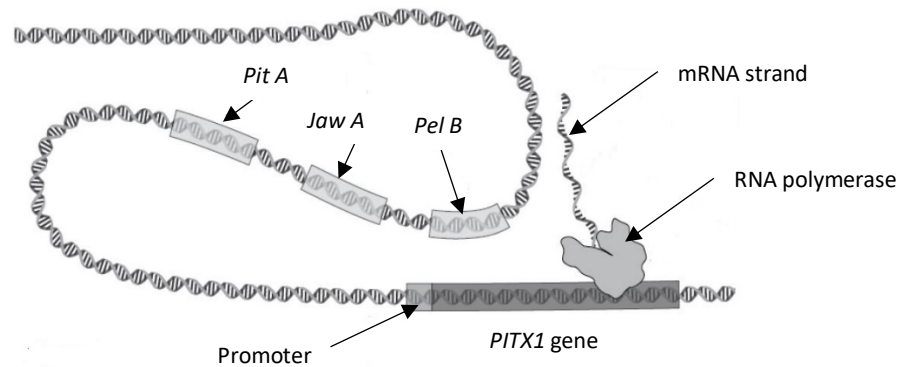


Fig. 5.1

- (i) The three regions, *Pel B*, *Jaw A*, *Pit A* are important for the expression of the *PITX1* gene.

Describe how the binding of an activator like *Pel B* at the *Pel B* region can affect gene expression. [2]

1. Binding of activator to enhancer causes **bending of spacer DNA** allows **direct interaction** of activators with **RNA polymerase** and **general transcription factors** at the promoter;
2. Promoting the assembly of **transcription initiation complex**, thus **increasing rate of transcription**;

- (ii) Explain how the structure of the *Pel B* activator enables it to bind to the *Pel B* region. [1]

1. *Pel B* activator protein has a **DNA-binding site** that is **complementary** to the **conformation** of the *Pel B* **enhancer / sequence**;

- (iii) Assuming that cells in the pelvis and eyes of the stickleback contain the same *PITX1* gene and the same three regions as shown in Fig. 5.1, explain why eyes of the stickleback do not have spines. [2]

1. In the **eye cells**, there are **no activator proteins** like *Pel B*, *Jaw A*, *Pit A* that can bind to the *Pel B*, *Jaw A* and *Pit A* enhancer regions;  
OR  
Genes for activator proteins *Pel B* / *Jaw A* / *Pit A* are silenced;  
OR  
Chromatin modification (e.g. DNA methylation)
2. Thus, the **PITX1 gene** in the cells of the eyes is **not expressed** and **no spines** are produced in the eyes;

- (c) Scientists researching on different species of sticklebacks living in the marine environments and in freshwater lakes found that they differed in the presence or absence of the pelvic spines as shown in Fig. 5.2.

Molecular genetic analysis found that these two groups of fishes exhibited very different levels of expression for the *PITX1* gene. Fig. 5.3 shows the level of expression (in arbitrary units or A.U.) of the *PITX1* gene in the pelvic regions for the fishes from both habitats.



Fig. 5.2

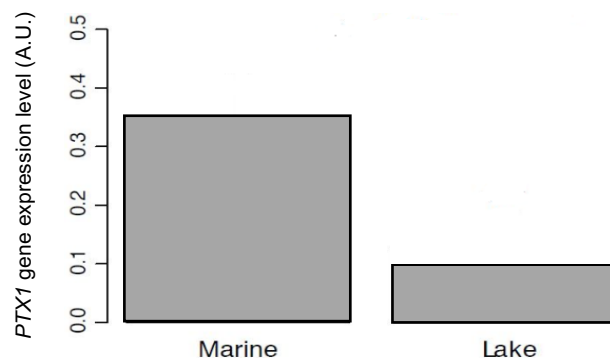


Fig. 5.3

- (i) With reference to Fig. 5.3, suggest a reason for the absence of pelvic spines in freshwater stickleback. [2]

1. Gene expression of **0.35 A.U.** in lake/freshwater fishes compared to **0.10 A.U.** in marine fishes; (A: 3.5 times gene expression level)
2. Not enough gene expression of *PITX1* gene, need to reach threshold/minimum level (OWTTE);

- (ii) In freshwater habitats, dragonfly larva are much larger in size than young sticklebacks. They predate on the young sticklebacks by grabbing on to protruding parts of the fish.

Suggest how the absence of pelvic spines aids in the survival of the young sticklebacks. [1]

1. It makes it **harder** for dragonfly larva to grab and eat the young fish due to **less** of protruding spines (OWTTE);

[Total: 10]

6 The patty pan squash plant, *Cucurbita pepo*, produces edible fruits that vary in colour.

(a) The colour of the fruits is controlled by two genes, **A/a** and **B/b**, that occur on different chromosomes.

- Allele **A** produces a white fruit colour.
- Allele **a** does not produce a colour by itself but allows the colours coded by gene **B/b** to show in the phenotype.
- Allele **B** produces a yellow fruit colour.
- Allele **b** produces a green fruit colour.

In a dihybrid cross, a pure-breeding white parent plant was crossed with another pure-breeding green parent plant. All the resulting F<sub>1</sub> plants produced white fruits.

The F<sub>1</sub> plants were then crossed with each other to obtain the F<sub>2</sub> generation.

(i) State the name for this type of gene interaction. [1]

Epistasis / Dominant epistasis;

(ii) Draw a genetic diagram to show the cross of the F<sub>1</sub> plants to obtain the F<sub>2</sub> generation.

|                      |                                                         |               |                                                         |                |                |
|----------------------|---------------------------------------------------------|---------------|---------------------------------------------------------|----------------|----------------|
| F <sub>1</sub>       | white squash                                            | X             | white squash                                            |                |                |
| Phenotypes           |                                                         |               |                                                         |                |                |
| F <sub>1</sub>       | AaBb                                                    | X             | AaBb                                                    |                |                |
| Genotypes            |                                                         |               |                                                         |                |                |
| Meiosis              |                                                         |               |                                                         |                |                |
| Gametes              | <div>AB</div> <div>Ab</div> <div>aB</div> <div>ab</div> |               | <div>AB</div> <div>Ab</div> <div>aB</div> <div>ab</div> |                |                |
| Random Fertilisation |                                                         |               |                                                         |                |                |
|                      | <div>AB</div>                                           | <div>Ab</div> | <div>aB</div>                                           | <div>ab</div>  |                |
|                      | <div>AB</div>                                           | AABB<br>white | AABb<br>white                                           | AaBB<br>white  | AaBb<br>white  |
|                      | <div>Ab</div>                                           | AABb<br>white | AAbb<br>white                                           | AaBb<br>white  | Aabb<br>white  |
|                      | <div>aB</div>                                           | AaBB<br>white | AaBb<br>white                                           | aaBB<br>yellow | aaBb<br>yellow |
|                      | <div>ab</div>                                           | AaBb<br>white | Aabb<br>white                                           | aaBb<br>yellow | aabb<br>green  |

F<sub>2</sub>  
Genotypic  
Ratio

12 A\_B\_ , A\_bb : 3 aaB\_ : 1 aabb

F<sub>2</sub> 12 white : 3 yellow : 1 green  
Phenotypic  
Ratio

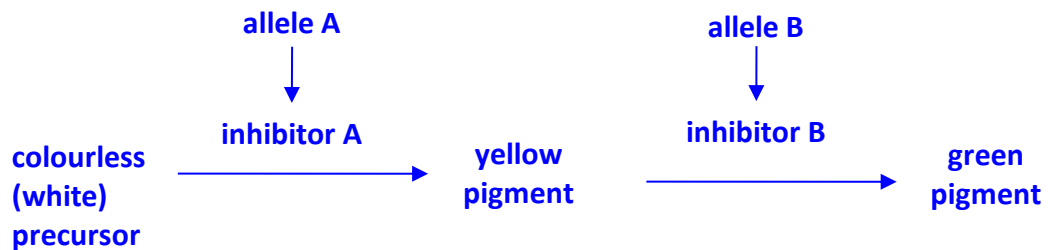
F<sub>1</sub> Phenotype **and** Genotype; 1 m  
Gametes; 1m (**R**: if not circled)  
Punnet Square with genotypes **and** phenotypes; 1m  
F<sub>2</sub> Genotypic and Phenotypic Ratios; 1 m

[4]

(b) Explain how different genotypes give rise to a white phenotype. [3]

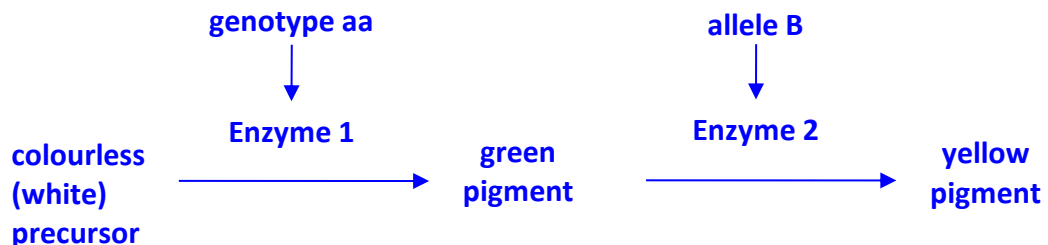
1. Both alleles A and B code for **functional inhibitors** A and B respectively;
2. Inhibitor A **inhibits** the enzyme that **converts a colourless/white precursor to yellow pigment**, while inhibitor B **inhibits** the enzyme that **converts the yellow pigment to a green pigment**;
3. Thus a genotype **with one dominant A allele or with at least one dominant A and at least one dominant B allele** will give a **white phenotype** ( $A\_bb$ ,  $A\_B\_$ );

Note: The squash is only **yellow** if the genotype is  $aaB\_$  / The squash is green if the genotype is  $aabb$  / OWTTE;



Accepted Alternative Answer:

1. **Allele A** will result in a **non-functional enzyme 1**;
  2. Non-functional enzyme 1 **cannot convert colourless/white precursor to green pigment**, hence a **white phenotype** ( $A\_bb$ ,  $A\_B\_$ );
  3. **Genotype aa** code for a **functional enzyme 1**, it converts **colourless/white precursor to green pigment** ( $aabb$ );
- (Note: Genotype of  $aaB\_$  will result in yellow phenotype)



(c) Explain how you would determine if the genotype of a white squash was homozygous dominant or homozygous recessive at gene locus **B**, assuming that it is heterozygous at gene locus **A**. [2]

1. **Test cross** with a plant that is/has **double homozygous recessive / genotype aabb / green** squash;
2. If we see **half** of the offspring are **white** and **half** of the offspring are **yellow**, the B locus is homozygous dominant,  $AaBB$ , but if **half white** and **half green**, then the B locus is homozygous recessive,  $Aabb$ ;

$AaBB \times aabb \rightarrow AaBb$  (white) :  $aaBb$  (yellow)  
 $Aabb \times aabb \rightarrow Aabb$  (white) :  $aabb$  (green)

[Total: 10]

- 7 (a) Chemiosmosis is the term used to describe the synthesis of ATP using a proton gradient across a membrane in a mitochondrion or chloroplast. It was first demonstrated by Peter Mitchell in 1961.

- In some of his experiments, Peter Mitchell used mitochondria that had been isolated from cells.
- The mitochondria were kept in liquid, in glass dishes, to which ADP, Pi and other substances were added.
- The outer mitochondrial membrane is freely permeable to proton entry.
- The temperature, pH and water potential were kept constant.
- After a period of time, he checked for the presence of ATP.

The contents of some of the dishes are shown in the table below.

Complete the table using a tick (✓) if ATP was produced, and a cross (✗) if no ATP was produced. [2]

| dish contents                                                             | ATP produced |
|---------------------------------------------------------------------------|--------------|
| mitochondria + ADP + Pi + acetyl CoA + oxygen                             | ✓            |
| mitochondria + ADP + Pi + acetyl CoA                                      | ✗            |
| mitochondria + ADP + Pi + low concentration of protons (H <sup>+</sup> )  | ✗            |
| mitochondria + ADP + Pi + high concentration of protons (H <sup>+</sup> ) | ✓            |

0 or 1 correct = 0 mark

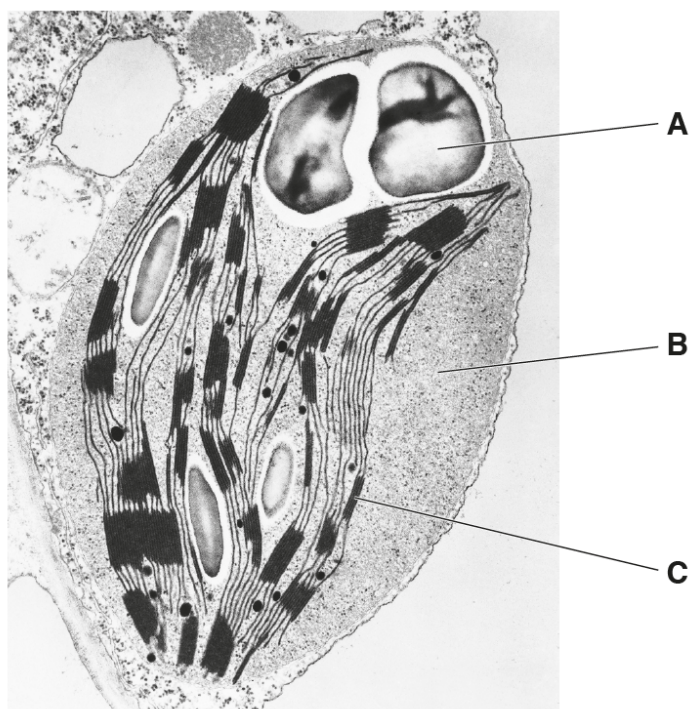
2 or 3 correct = 1 mark

4 correct = 2 marks

- (b) Explain the consequences to a mitochondrion if the water potential of the liquid in the dishes is higher than the water potential of the mitochondrial matrix. [2]

1. Water would enter the mitochondrion via osmosis down the water potential gradient;
2. **Mitochondrion burst / membranes rupture**;

(c) Fig. 7.1 is a transmission electron micrograph of a chloroplast.



**Fig. 7.1**

Many compounds and structures involved in photosynthesis are located in a chloroplast.

Using the labels **A**, **B** or **C**, complete Table 7.1 to show the location of four of these compounds or structures.

You may use each of the letters **A**, **B** and **C** once, more than once, or not at all.

**Table 7.1**

| compound or structure | location |
|-----------------------|----------|
| ATP synthase          | <b>C</b> |
| rubisco               | <b>B</b> |
| starch grain          | <b>A</b> |
| phospholipid bilayer  | <b>C</b> |

*0 or 1 correct = 0 mark*

*2 or 3 correct = 1 mark*

*4 correct = 2 marks*

[2]

- (d) The light dependent stage of photosynthesis in a suspension of isolated chloroplasts can be investigated using the Hill reaction.

Dichlorophenolindophenol (DCPIP) can be used to follow the process. DCPIP is a blue dye which turns colourless when it is reduced by accepting hydrogen and electrons.

oxidised DCPIP  $\longrightarrow$  reduced DCPIP

blue  $\longrightarrow$  colourless

- (i) DCPIP is an artificial hydrogen acceptor that can be used in the Hill reaction.

Name the natural hydrogen acceptor found in chloroplasts that is replaced by DCPIP in the Hill reaction. [1]

**NADP<sup>+</sup>**

- (ii) Outline the way in which hydrogen is made available to reduce the hydrogen acceptor in the light dependent stage of photosynthesis. [2]

1. **Photolysis** of **water** occurs to **split water molecule into hydrogen ions and oxygen atom** in **photosystem II**
2. NADP reductase / enzyme **combines NADP<sup>+</sup> with hydrogen ions and electrons to reduce it to NADPH.**

- (iii) Predict **and** explain the effect on the concentration of RuBP in the chloroplasts if DCPIP becomes reduced instead of the natural hydrogen acceptor. [1]

1. RuBP concentration will **decrease** as it is **used up in carbon fixation** but **not regenerated.**

[Total: 10]

8 Sickle cell anaemia is an autosomal recessive inherited disorder.

- The Hb<sup>A</sup> allele codes for the normal  $\beta$ -globin polypeptide of haemoglobin.
- The Hb<sup>S</sup> allele, caused by a base substitution mutation, codes for an abnormal  $\beta$ -globin polypeptide.
- The base substitution results in the amino acid glutamate, which has a charged R group, to be replaced by valine, which has a non-polar R group, in the polypeptide.

The abnormal haemoglobin molecules (HbS) form fibres in low partial pressures of oxygen ( $pO_2$ ). The fibres cause red blood cells to become sickle shaped and the cells can block blood capillaries.

Individuals with adult haemoglobin molecules that are all abnormal (HbS) have sickle cell anaemia. This is a painful chronic condition that can be life-threatening.

(a) Explain why this mutation causes the HbS to form fibres. [2]

1. The **R group** of **valine** is **non-polar** and **hydrophobic** while the R group of glutamate is charged and hydrophilic. (no need 2<sup>nd</sup> part because not about HOW)
2. **HbS polymerise** to form fibres due to **hydrophobic interactions** between exposed hydrophobic regions on  $\beta$ -globin polypeptides of adjacent haemoglobin molecules (**R: without idea of between haemoglobin**) in low partial pressures of oxygen ( $pO_2$ )

(b) People who are heterozygous (Hb<sup>A</sup> Hb<sup>S</sup>) have sickle cell trait (SCT). For a child to inherit sickle cell anaemia (Hb<sup>S</sup> Hb<sup>S</sup>), both parents must have SCT. A genetic screening program is available for sickle cell anaemia and SCT.

- (i) To test for the presence of Hb<sup>S</sup>, DNA is extracted and the polymerase chain reaction (PCR) is carried out with two specific sets of primers. One set of primers (normal-specific primers) detects Hb<sup>A</sup> allele while the other set of primers (mutant-specific primers) detects Hb<sup>S</sup> allele.

Explain:

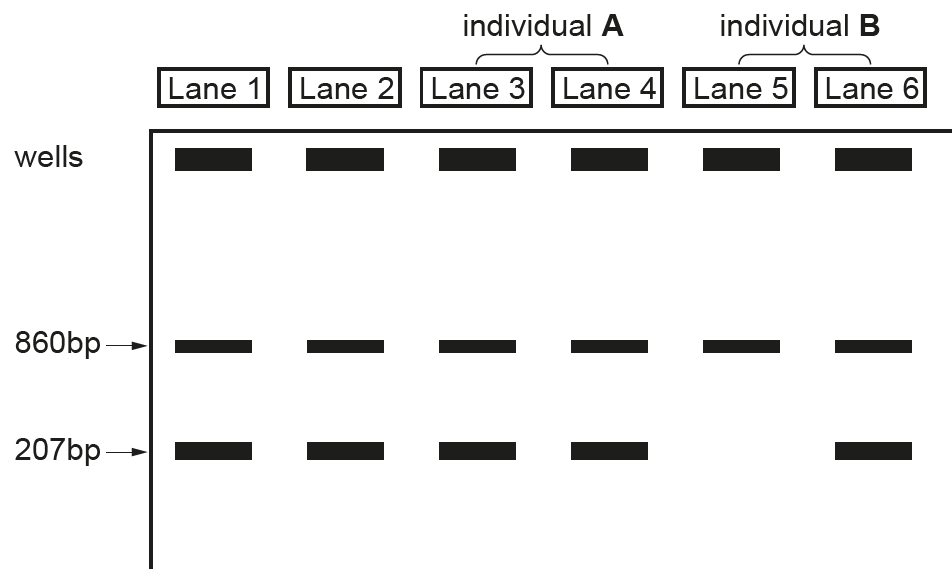
- why primers are used in PCR
- how the use of two specific sets of primers allows the amplification of the normal, sickle cell anaemia and SCT genotypes. [2]

1. Primers **anneal/bind to normal or mutated codon to provide free 3' OH group for Taq polymerase to extend/replicate/amplify DNA;**
2. Binding of **normal-specific primer only** indicates **Hb<sup>A</sup> Hb<sup>A</sup>** (normal genotype), binding of **mutant-specific primer only** indicates **Hb<sup>S</sup> Hb<sup>S</sup>** (sickle cell anaemia genotype), while binding of **both normal-specific and mutant-specific primers** indicate **Hb<sup>A</sup> Hb<sup>S</sup>** (SCT genotype)

(ii) Gel electrophoresis is carried out on the products of the PCRs.

Fig. 8.1 includes the results for two individuals, **A** and **B**, tested for the sickle cell allele.

- Each lane has an 860 base pair (bp) band to indicate the test is valid.
- Lane 1 is a control lane with a 207bp band for an individual with known normal phenotype.
- Lane 2 is a control lane with a 207bp band for an individual with known sickle cell anaemia phenotype.
- Lanes 1, 3 and 5 contain DNA from the PCR that used normal-specific set of primers.
- Lanes 2, 4 and 6 contain DNA from the PCR that used mutant-specific set of primers.



**Fig. 8.1**

Deduce the genotypes and phenotypes of individuals **A** and **B** in Fig. 4.1. [2]

**A: Hb<sup>A</sup> Hb<sup>S</sup> SCT**

**B: Hb<sup>S</sup> Hb<sup>S</sup> Sickle cell anaemia**

(c) Genetically modified stem cell transplant is a common form of gene therapy used to treat sickle cell anaemia. It involves the following steps:

1. Removing haematopoietic stem cells (HSCs) from the patient
2. Genetically modifying the genome of the HSCs by inserting normal Hb<sup>A</sup> allele
3. Reintroducing the modified stem cells back into the patient.

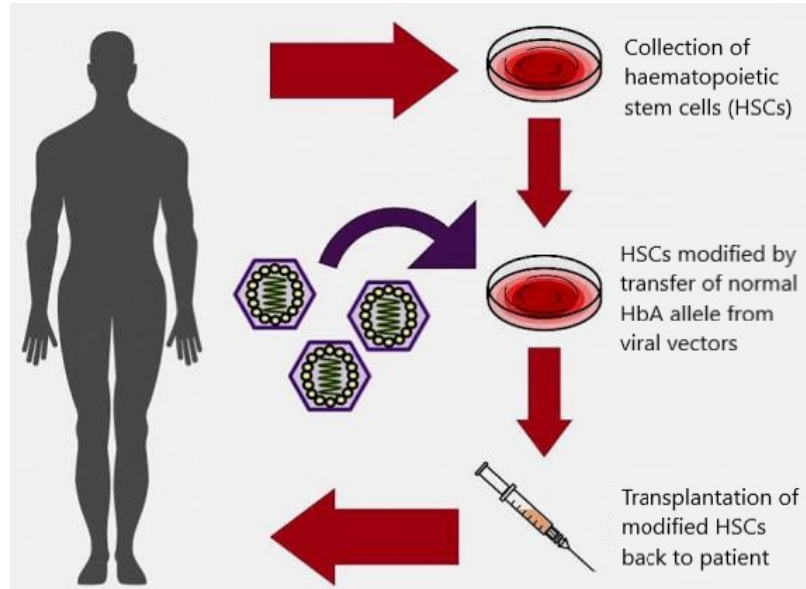


Fig. 8.2

(i) Describe the characteristics of haematopoietic stem cells that enable them to be used for stem cell therapy. [3]

1. Haematopoietic stem cells are **multipotent** with the ability to **differentiate** into **several blood cell types**. (several related cell types but is restricted to blood cells only);
2. Hence, they are able to **replace sickled red blood cells** (R: diseased tissues) (or **replenish blood cells lost through disease**);
3. They are capable of **self-renewal** and **extensive proliferation** to form more haematopoietic stem cells;

(ii) Suggest an advantage of the specific type of stem cell therapy shown in Fig. 8.2. [1]

1. As the haematopoietic stem cells are taken from the patient's own body, the cells are **genetically identical** to that of the patient, **avoiding the problem of immune rejection** (or lower chance);
2. AVP (no need to wait for stem cells from healthy donor, no need to find a genetically-compatible stem cell donor)

[Total: 10]

- 9 Different signalling pathways generate calcium ( $\text{Ca}^{2+}$ ) signals which regulate many cellular functions, e.g. smooth muscle contraction.

Fig. 9.1 shows the inositol trisphosphate/calcium ( $\text{IP}_3/\text{Ca}^{2+}$ ) signalling pathway. Phosphatidylinositol-bisphosphate ( $\text{PIP}_2$ ) is hydrolysed into  $\text{IP}_3$  and diacylglycerol (DAG) by phospholipase C.

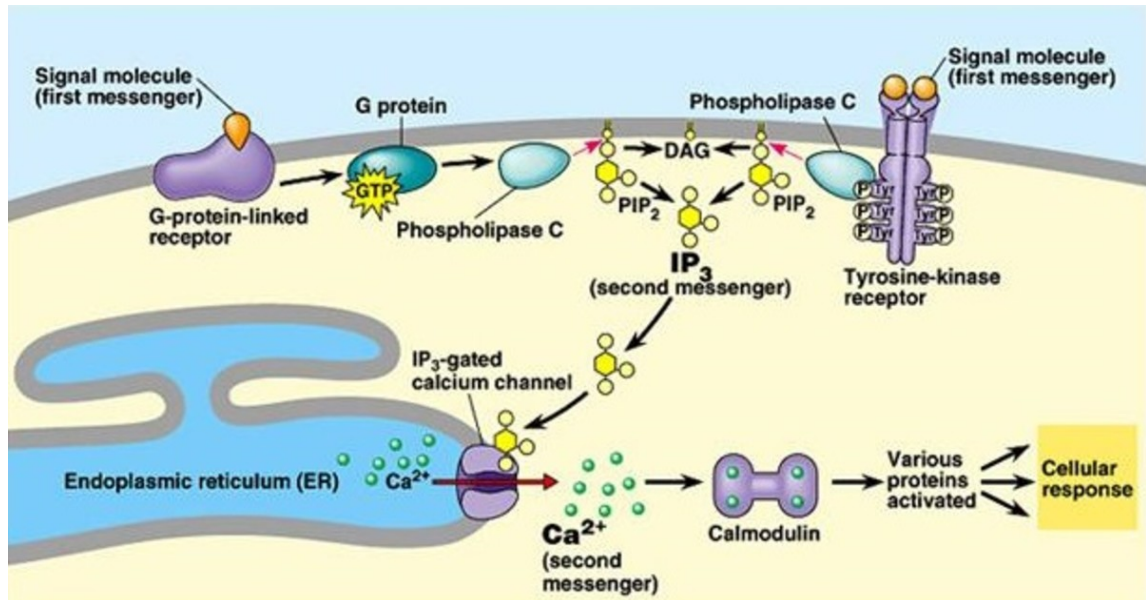


Fig. 9.1

(a) With reference to Fig. 9.1,

- (i) describe the structural differences of the two receptors. [2]
  1. **One protein vs two protein subunits** (dimer)
  2. **Different ligand-binding sites** complementary in **conformation** (and charge) to different ligands (**A: different conformation of ligand-binding sites**)
  3. **Different binding sites for G protein vs phospholipase C** in intracellular domain
- (ii) define second messengers and explain their role in signal transduction. [2]

define: Small, non-protein, water-soluble molecules/ions

role: Bind, induce change in conformation and activate ...

(e.g.  $\text{IP}_3$  binds and induces change in conformation in  $\text{IP}_3$ -gated calcium channel, activating the  $\text{IP}_3$ -gated calcium channel

OR  $\text{Ca}^{2+}$  binds and induces change in conformation in calmodulin, activating the calmodulin)

- (iii) outline two stages where signal amplification may occur. [2]
1. Each tyrosine kinase receptor activating many phospholipase C, where each binds to a phosphorylated tyrosine residue
  2. Each phospholipase C producing many IP<sub>3</sub>
  3. Each IP<sub>3</sub> activating a IP<sub>3</sub>-gated calcium channel and releasing many Ca<sup>2+</sup>
- (iv) Dysregulation of the IP<sub>3</sub>/Ca<sup>2+</sup> signalling pathway can lead to many different possible human diseases. Hypertension is caused by increased smooth muscle contraction due to enhanced IP<sub>3</sub>/Ca<sup>2+</sup> signalling.

Suggest the type of mutation in phospholipase C that could lead to hypertension. [2]

1. Gain-in-function mutation in gene coding for phospholipase C, causing phospholipase C to be constitutively active / have high rate of hydrolysis of PIP<sub>2</sub>
  2. Resulting in constitutive signal transduction, thus high concentration of Ca<sup>2+</sup> leading to increasing smooth muscle contraction / hypertension
- (b) Calcium binding by calmodulin exhibits considerable cooperativity, making calmodulin an unusual example of a monomeric (single-chain) cooperative-binding protein with four calcium-binding sites.

Fig. 9.2 shows calmodulin without calcium (left), and calmodulin with calcium (right). Sites that bind target proteins are indicated by the stars (\*).

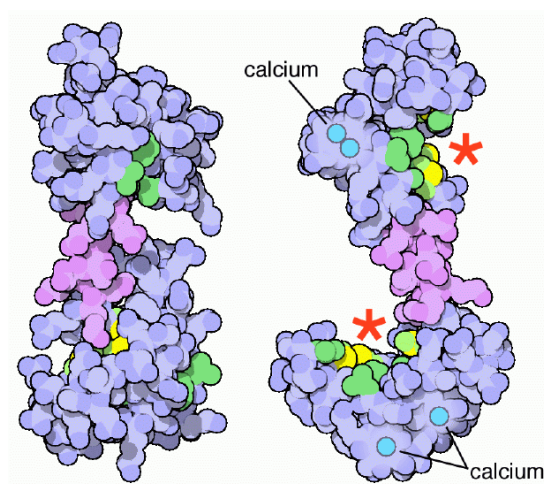


Fig. 9.2

With reference to Fig. 9.2, suggest how cooperative binding of Ca<sup>2+</sup> ions to calmodulin is necessary for calmodulin activity. [2]

1. [Cooperative binding] Binding of one Ca<sup>2+</sup> ion to one calcium-binding site of calmodulin induces conformational change in the other three calcium-binding sites.
2. This leads to an increase in the affinity of calcium-binding sites for Ca<sup>2+</sup> and causes the binding site for target protein to be exposed.

[Total: 10]

10 (a) Fig. 10.1 is a diagram that shows the structure of an antibody molecule.

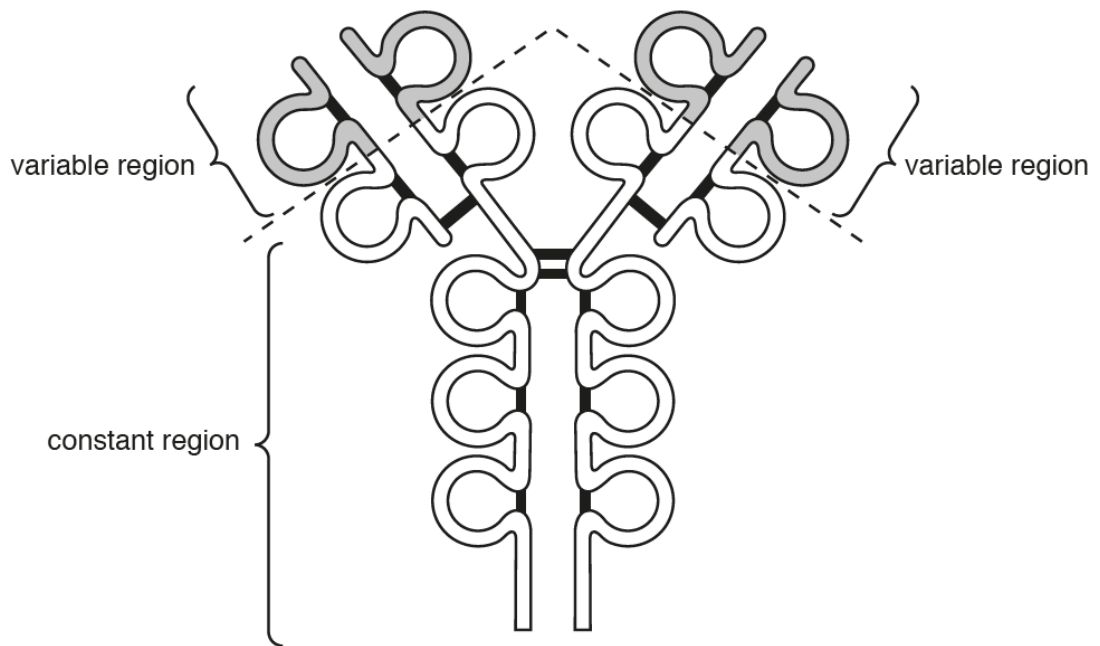


Fig. 10.1

Use Fig. 10.1 to explain how the structure of the variable region of an antibody molecule is related to its function. [2]

1. It has two antigen-binding sites that are complementary in conformation and charge to (epitope of) antigen; (R: active site)
2. Both light chain and heavy chain (polypeptide chain) undergo folding (A: intramolecular interactions between R groups of amino acids) to give rise to the specific 3D conformation;
3. Preventing pathogen from binding to host cell receptors and infecting host cells, thus neutralizing pathogen.

(b) Fig. 10.2 shows a process that occurs to heavy chain gene on human chromosome 14.

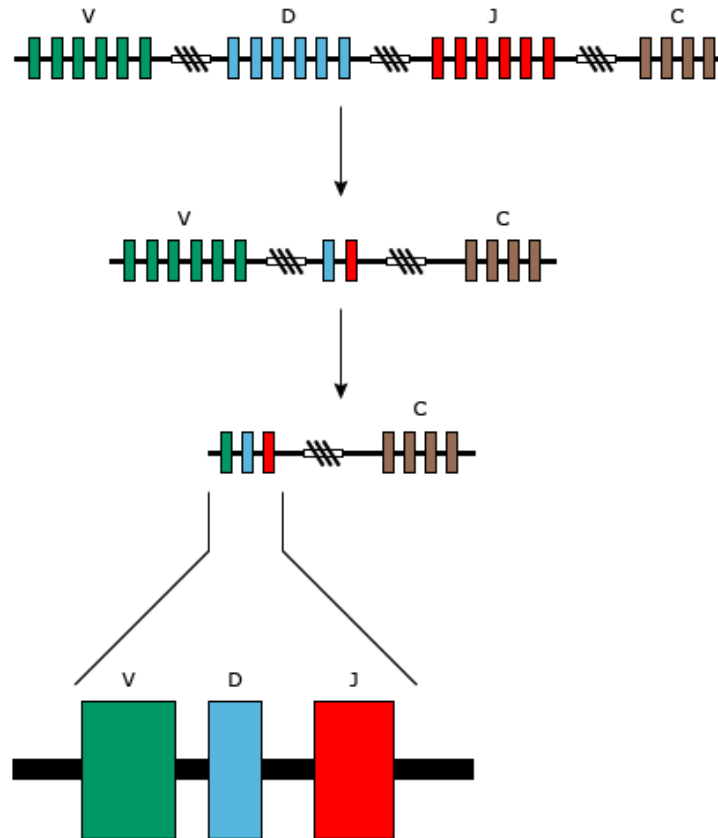


Fig. 10.2

- (i) Outline the process shown in Fig. 10.2. [2]
1. **Rearrangement** of a D gene **segment** with a J gene **segment** to form a **DJ rearrangement**
  2. **Rearrangement** of a V gene **segment** with the DJ rearrangement to form **VDJ rearrangement**

(R: plural)

- (ii) Explain the significance of the above process. [1]

1. Help to generate a **variety** of **heavy chain** of antibodies after transcription, splicing and translation

[Total: 5]

- 11 Fig. 11.1 shows the relationship between the density of coral skeletons and the rate of precipitation of key minerals for the coral skeleton. The different shapes represent coral samples obtained from different locations in the world.

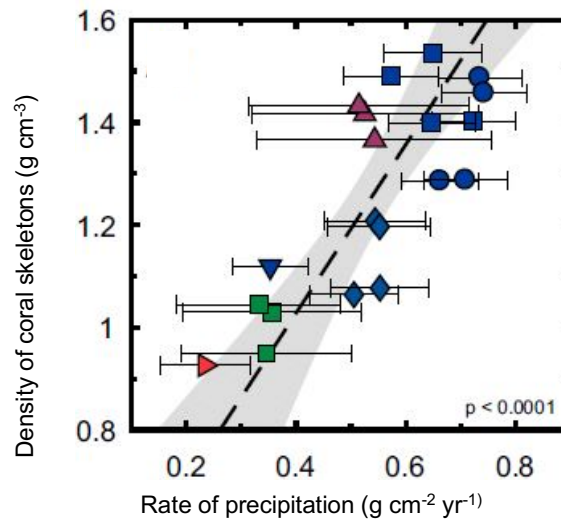


Fig. 11.1

- (a) (i) Describe the general relationship between density of coral skeletons and rate of precipitation of key minerals for the coral skeleton. [1]

1. Increased rate of precipitation correlates / directly proportional to the increase in density of coral skeletons

- (ii) An increase in carbon dioxide concentration in the atmosphere has resulted in higher concentrations of carbon dioxide in the ocean. This has caused a decrease in the pH of the ocean and has resulted in ocean acidification.

Explain how ocean acidification has affected corals. [1]

1. Ocean acidification has reduced quantity of carbonate ions / calcium carbonate in the oceans that are required for building corals leading to lower density of the coral skeleton and slower growth / slower rebuilding;

- (b) Scientists are studying seaweeds such as *Laminaria hyperborea* because they absorb a large quantity of carbon dioxide during photosynthesis. This may help to increase the pH of the ocean and reverse ocean acidification.

In the laboratory, *Laminaria hyperborea* was grown in water with different pH values. All other variables, including temperature and light, were standardised.

The mean rate of photosynthesis was calculated over a 24 hour period for each pH value.

The results are shown in Fig. 11.2.

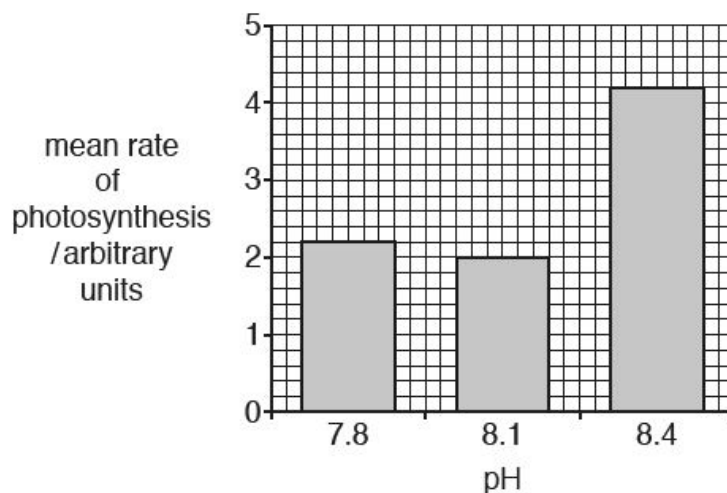


Fig. 11.2

The lower pH values on Fig. 11.2 represent ocean acidification.

With reference to Fig. 11.2, suggest why the results for the lower pH values do **not** fully support the idea that seaweeds can help to reduce ocean acidification. [3]

1. At lower pH of 7.8 / 8.1, mean rate of photosynthesis was 2.2 units / 2.0 units;
2. This is 2.0 units / 2.2 units lower/less compared to rate of photosynthesis at pH 8.4 / OWTTE
3. Thus the seaweeds can absorb less carbon dioxide at lower pH, which does not fully support idea they can help reduce ocean acidification;

[Total: 5]

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